Physics II - Electromagnetism - 25 June 2026 - Part A: 15 multiple choice (30 min), Part B: problems (90 min)

Q1. Two point charges q1 = 2 uC and q2 = -3 uC are 0.5 m apart. Find the force.

Q2. A long straight wire carries I = 5 A. Find B at 10 cm using Ampere's law.

Problem 1 (10 pts). A conducting sphere of radius R carries charge Q. Find E(r) everywhere using Gauss's law and the potential V(r).

Problem 2 (10 pts). A rectangular loop of resistance R, with side L along the field boundary, moves to the right out of a uniform magnetic field B pointing into the page, at constant speed v. Find the induced emf, the current and its direction (Lenz).